

भारतीय मानक
Indian Standard

IS 2112 : 2025

**चाँदी और चाँदी मिश्रधातु,
आभूषण/शिल्पकारी — शुद्धता और
मुहरांकन — विशिष्टि**
(चौथा पुनरीक्षण)

**Silver and Silver Alloys,
Jewellery/Artefacts — Fineness and
Marking — Specification**
(Fourth Revision)

ICS 39.060

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March 2025

Price Group 4

Precious Metals Sectional Committee, MTD 10

FOREWORD

This Indian Standard (Fourth Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Precious Metals Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was first published in 1962, and later revised in 1981, 2003 and 2014.

In this revision, following major changes have been done:

- a) All amendments issued the previous version have been incorporated;
- b) Bullion and coin have been defined and included in the scope of the standard. In the note to scope it is made clear that all articles of silver are covered under artefacts;
- c) Silver jewellery/artefacts grades of fineness 900 has been deleted and a new grade of fineness 958 has been added, in line with ISO 9202; and
- d) Marking [7](#) has been modified.

The composition of the Committee responsible for the formulation of this standard is given in [Annex A](#).

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded of value should be the same as that of the specified in this standard.

Indian Standard

SILVER AND SILVER ALLOYS, JEWELLERY/ARTEFACTS — FINENESS AND MARKING — SPECIFICATION

(Fourth Revision)

1 SCOPE

1.1 This standard specifies the grades of silver/silver alloys based on silver content:

- Three grades of fine silver used in manufacture of bullion and coins; and
- Seven grades of silver alloys used in the manufacture of jewellery/artefacts.

1.2 This standard also specifies the requirement for marking of fineness and other details to be marked on tested silver bullion and jewellery/artefact.

NOTE — Artefacts cover medallions, utensils and any article of silver.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards:

IS No.	Title
IS 1387 : 1993	General requirements for the supply of metallurgical materials (<i>second revision</i>)
IS 2113 : 2014	Assaying silver in silver and silver alloys — Methods (<i>third revision</i>)
IS 15820 : 2024	Establishment and operation of assaying and hallmarking centres — General requirements (<i>first revision</i>)
IS 16900 : 2021/ ISO 15096 : 2020	Jewellery and precious metals — Determination of high purity silver — Difference method using ICP-OES (<i>first revision</i>)

3 TERMINOLOGY

For the purpose of this standard, the following

definitions shall apply.

3.1 Assaying — The method of accurate determination of silver content of the sample expressed in parts per thousand (‰).

3.2 Bullion — Refined bars of various shapes of fine silver manufactured by mints/refineries.

3.3 Coin — Silver coins of fineness 999.9, 999.5 and 999.0 manufactured by mints/refineries.

3.4 Fineness — The ratio between the mass of silver content and total mass expressed in parts per thousand (‰).

3.5 Hallmark — Hallmark means in relation to precious metal article, the Standard Mark, which indicates the proportionate content of precious metal in that article as per the relevant standard.

4 SUPPLY OF MATERIAL

General requirements relating to the supply of material are laid down in IS 1387.

5 GRADES

Fine silver and silver alloys shall be classified in accordance with the following grades depending upon their fineness:

Sl No. (1)	Grade (2)		Fineness (3)
i)	9 999 9 995 999	Fine silver for bullion and coins	999.9 999.5 999.0
ii)	999	Fine silver for jewellery artefacts	999.0
	990 970 958 925 835 800	Silver alloys for jewellery artefacts	990.0 970.0 958.0 925.0 835.0 800.0

NOTE — All the above grades of silver and silver alloys shall be free from cadmium and lead.

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6 REQUIREMENTS

6.1 Fine silver for bullion and coins and silver alloys used for manufacture of jewellery/artefacts shall have the fineness as given in [5](#) without any negative tolerance.

6.2 Cadmium and lead shall not be present in silver jewellery/artefacts.

NOTE — In XRF analysis of silver jewellery/artefacts, the value of cadmium and lead below 0.01 percent may be considered as absent.

6.3 In the manufacture of jewellery/artefacts of grades 925, 958, 970 and 990, the solder of purity 750 silver minimum and for grade of 800 and 835, the solder of purity 650 silver minimum free from cadmium and lead shall be used. However, melting purity of articles shall not be less than the purity for which it has been offered for hallmarking as per the grades given in [5](#).

6.4 Sampling

Unless otherwise specified for the jewellery/artefacts of the same design and quality, the sample shall be taken from 10 percent of the semi-finished/finished jewellery/artefacts. Any other sampling plan may also be followed subject to the agreement between the contracting parties or as specified by the inspection agency as the case may be. The sampling plan mentioned in IS 15820 may also be referred.

The sampling of bullion and coins shall be done as specified in IS 2113/inspection agency.

6.5 Assaying

6.5.1 The sample taken as prescribed in [6.4](#) shall be assayed in accordance with IS 2113. The actual weight of samples, etc, shall however, be as prescribed in respective test methods as detailed in IS 2113.

6.5.2 For determination of fineness of fine silver IS 16900 is preferred method.

6.5.3 Detection of cadmium and lead in silver and silver alloys shall be carried out by X-ray fluorescence (XRF) method.

6.6 Tolerance

No negative tolerance shall be allowed on the silver content of silver bullion, coin, and jewellery artefact sampled and tested as per [6.4](#) and [6.5](#) respectively.

7 MARKING

7.1 The silver and silver alloys, jewellery/artefacts shall be hallmarked by laser marking by BIS recognized assaying and hallmarking centre only. The hallmarked articles shall carry the following markings:

- BIS Standard Mark with the word SILVER;
- Grade; and
- 'Unique alphanumeric ID' of the article issued by the assaying and hallmarking centre'.

7.2 The silver bullion and silver coins shall be hallmarked by punching or embossing by BIS recognized refineries/mints only. The hallmarked articles shall carry the following markings:

- BIS Standard Mark with the word silver;
- Grade;
- Name/Identification of the manufacturer; and
- Serial number.

NOTE — In case of coins, serial numbers may be marked on the packaging.

7.3 Hallmarking

The use of Hallmark is governed by the provisions of the *Bureau of Indian Standards Act, 2016* and Rules and Regulations made thereunder. The details of conditions under which the registration for the use of the Hallmark may be granted to jewellers or jeweller/artefact manufacturers may be obtained from the Bureau of Indian Standards.

7.3.1 Marking should be preferably linear.

7.3.2 Markings shall be legible and durable. Marking shall be visible by a magnifying glass of magnification 10 X.

7.3.3 Markings shall be done on all parts which can be easily removed or replaced.

7.3.4 Silver jewellery/artefacts on which marking should not be applied are as follows:

- Any silver jewellery/artefact which has not been assayed by BIS recognized assaying and hallmarking centre and does not confirm to the fineness/grade declared;

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- b) Any silver jewellery/artefact which fails to comply with specified fineness;
 - c) Any silver jewellery/artefact which is not solid or which is made with hollow centre and then filled with base metal, cement, lac or other substance thereby giving the article a false or added weight or strength unless
 - d) Silver jewellery/artefact on which it is physically not possible to stamp the marks.
- the weight of silver is declared by the jeweller and marked on the article by the centre; and

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ANNEX A

(Foreword)

COMMITTEE COMPOSITION

Precious Metals Sectional Committee, MTD 10

<i>Organization</i>	<i>Representative(s)</i>
Geological Survey of India, New Delhi	SHRI PARAVJEET SINGH (Chairperson)
All India Gems and Jewellery Trade Federation, Mumbai	SHRI D. D. KAREL SHRI SURESH I DHARUV (<i>Alternate</i>)
Association of Gold Refineries and Mints, New Delhi	SHRI ANIL KANSARA SHRI ANIL C. KANSARA (<i>Alternate</i>)
Bhartiya Swarnkar Sangh, Jaipur	SHRI DULI CHAND KAREL SHRI SURATN KUMAR MANISH (<i>Alternate</i>)
Consumer Guidance Society of India, Mumbai	SHRI SITARAM DIXIT SHRI M. S. KRAMATH (<i>Alternate</i>)
CGR Metalloys Private Limited, Kochi	SHRI JAMES JOSE SHRI JOSEPH K. JAMES (<i>Alternate</i>)
CSIR - National Chemical Laboratory, Pune	DR SAKYA SEN DR K. KRISHNAMOORTHY (<i>Alternate</i>)
CSIR - National Physical Laboratory, New Delhi	DR S. P. SINGH SHRI DINESH SINGH (<i>Alternate</i>)
Gem and Jewellery Export Promotion Council, Mumbai	SHRI SABYASACHI RAY DR NAWAL KISHOR AGARWAL (<i>Alternate</i>)
Hindalco Industries Limited, Mumbai	SHRI JAYESH PAWAR SHRI DIVYANG SHAH (<i>Alternate</i>)
India Government Mint, Mumbai	SHRI BIMAL PARSAD DHAL SHRI RAVINDRA GUNDERAO JADHAV (<i>Alternate</i>)
Indian Association of Hallmarking Centres, New Delhi	SHRI UDAY SHINDE SHRI ABHISHEK NIKAM (<i>Alternate</i>)
Indian Bullion & Jewellers Association Limited, Mumbai	SHRI CHETAN BHANDARI SHRI SURENDRA MEHTA (<i>Alternate</i>)
Indian Diamond Institute, Surat	SHRI SAMIR D. JOSHI SHRI HITESH VERMA (<i>Alternate</i>)
Institute of Chemical Technology, Mumbai	DR B. M. BHANAGE SHRI RADHE V. JAYARAM (<i>Alternate</i>)
Jalan and Company, Chandni Chowk, New Delhi	SHRI ISHWAR. JALAN SHRI VINAY JALAN (<i>Alternate</i>)
MMTC-PAMP India Private Limited, New Delhi	SHRI PANKAJ DESHMUKH SHRI ANKUR GOYAL (<i>Alternate</i>)
National Mineral Development Corporation, Hyderabad	SHRI R. K. GARG SHRI NIKHIL KUMAR (<i>Alternate</i>)
Sigma Four, New Delhi	SHRI A. K. BAHL SHRIMATI ANITA BHATIA (<i>Alternate</i>)

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<i>Organization</i>	<i>Representative(s)</i>
Titan Company Limited, Bengaluru	SHRIMATI VIJAYAKUMAR T. SHRI PRABU P. (<i>Alternate</i>)
Voluntary Organisation in Interest of Consumer Education (VOICE), New Delhi	SHRI B. K. MUKHOPADHYAY SHRI M. A. U. KHAN (<i>Alternate</i>)
World Gold Council, New Delhi	SHRI P. R. SOMASUNDARAM SHRI AJIT MAUSKAR (<i>Alternate</i>)
BIS Directorate General	SHRI SANJIV MAINI, SCIENTIST 'F'/SENIOR DIRECTOR AND HEAD (METALLURGICAL ENGINEERING) [REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>)]

Member Secretary
SHRI AJAY KUMAR SONI
SCIENTIST 'C'/DEPUTY DIRECTOR
(METALLURGICAL ENGINEERING), BIS

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